



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 3 • STANDARD 6.AT.2

Understand Rates and Unit Rates

Lesson 3-2 • Teacher Copy

Name: _____

Date: _____

Class: _____

ARCADE BUILDER MISSION

Best Booth in the Arcade

You are the new operations manager at NeonPlex Arcade. Two ticket booths are competing for players — Booth A charges \$3 for 5 games, Booth B charges \$5 for 8 games. Players want the best deal, and your job is to crown the winning booth by mastering unit rates.

My Notes

Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can find a unit rate to compare prices and decide the better buy.

Language Objective: I can explain my choice using the words rate, unit rate, per, and better buy.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Rates and Unit Rates

Rate

Unit Rate

Per

Ratio

Unit price

Better Buy

Divide



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

A rate compares different units; a
 compares a quantity to exactly 1 unit.

2

A ratio that compares two quantities with different units is a
.

3

A rate that tells the cost or amount for one unit is a
.

4

The word that means 'for each one' is .

5

A comparison of two quantities is a .

6 A unit rate that tells the cost of one item is the .

7 The option with the lower unit rate, costing less per item, is the .

8 To find a unit rate, I the price by the number of units.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Which size is the better deal, and how do you know?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐ **Rates and Unit Rates** — A rate compares different units; a unit rate compares a quantity to 1 unit.
Tasas y tasas unitarias — Una tasa compara unidades diferentes; una tasa unitaria compara una cantidad con 1 unidad.

☐ **Rate** — A ratio comparing two amounts with different units, like miles per hour.
Tasa — Una razón que compara dos cantidades con unidades distintas, como millas por hora.

☐ **Unit Rate** — A rate for just 1 of something, like cost for 1 item.
Tasa unitaria — Una tasa para solo 1 de algo, como el precio de 1 artículo.

☐ **Per** — For each one. Example: 5 dollars per book.
Por — Por cada uno. Ejemplo: 5 dólares por libro.

☐ **Ratio** — A way to compare two amounts, like 3 to 2.
Razón — Una manera de comparar dos cantidades, como 3 a 2.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The ____ bag is the better deal because its unit rate is \$ ____ per cup, which is ____ than \$ ____ per cup for the other size.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: It is hard to compare because the booths have different numbers of games.

Evidence: Students notice the prices cannot be compared directly because the number of games differs, so they need the cost for one game (the unit rate) to compare fairly.

Reasoning: This evidence connects the definition of rates and unit rates to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The ____ bag is the better deal because its unit rate is \$ ____ per cup, which is ____ than \$ ____ per cup for the other size.
- My answer is ____.

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.
- I know because ____.

Mi afirmación es ____.

Lo sé porque ____.

- So ____.

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Rates and Unit Rates
2. **Notes 2:** Rate
3. **Notes 3:** Unit Rate
4. **Notes 4:** Per
5. **Notes 5:** Ratio
6. **Notes 6:** Unit price
7. **Notes 7:** Better Buy
8. **Notes 8:** Divide
9. **Watch for this mistake:** *A common mistake in Rates and Unit Rates is comparing total costs instead of unit rates: a student sees Pack X at \$4.00 for 10 tokens and Pack Y at \$6.00 for 16 tokens and picks Pack X just because \$4.00 is the smaller number, without dividing to find the cost per token (\$0.40 for Pack X vs. \$0.375 for Pack Y — Pack Y is actually the better buy). A second common mistake is dividing backwards — finding $\text{tokens} \div \text{cost}$ instead of $\text{cost} \div \text{tokens}$ — which turns 'dollars per token' into 'tokens per dollar' and gives an answer that looks reasonable but answers the wrong question. Always divide the total cost by the number of items, and check that your unit rate answers 'the cost for ONE item,' not the reverse.*
10. **Try It 1:** A. Pulse, at 15 points per minute — Nova: $78 \div 6 = 13$ points per minute. Pulse: $75 \div 5 = 15$ points per minute. $15 > 13$, so Pulse is faster.
11. **Try It 2:** A. Machine B, at 21 coins per play — Machine A: $150 \div 5 = 30$ coins per play. Machine B: $168 \div 8 = 21$ coins per play. $21 < 30$, so Machine B is the better value.